The text and figures in this PDF are approved by the author(s) for publication. Any mistakes in this PDF will not be corrected by the publisher. This PDF was created on October 24, 2025.

Multiple probes and feature deletion in the Tira agreement complex

Himidan Hassen and Peter Jenks

1. Introduction

In this paper we show that the exponents of agreement in Tira are the realization of two separate probes, one on C, which targets DP topics, and another on T, sensitive to just pronouns. Obligatory T-to-C movement results in the realization of both probes on a single complex head. We show that two separate morphological deletion operations conceal the faithful realization of the φ-features copied by these probes. The first deletes the features on T when they are identical to those on C, an instance of Kinyalolo's Constraint (Kinyalolo 1991 a.o.). The second is characterized by the preferential realization of object features over subject features, an apparent instance of *Expone Outermost* (Deal 2022b, Grishin to appear). As Expone Outermost effects are found only with certain feature combinations in Tira, we analyze them as a special case of order-sensitive impoverishment.

2. Two sources of agreement in Tira finite clauses

Tira is a Kordofanian language (Niger-Congo). These languages are autochthonous to the Nuba Mountains of Sudan, though they are spoken by a large displaced community in Egypt, Kenya, and beyond. The coauthors of this paper have been collaborating via Zoom since 2020 with colleagues at UCSD, including Nina Hagen Kaldhol, Sharon Rose, and Mark Simmons. All of the data in this paper comes from the first author, who speaks the Kadar dialect of Tira. He spent his childhood in Komo village in the Nuba Mountains to the east of Kauda before moving to Kenya and then Canada. Recent publications on Tira include Hassen et al. (2022), Kaldhol (2024), Rose (2025), and Kaldhol et al. (2025).

Main clauses in Tira are characterized by V2 syntax. Every sentence has an initial constituent immediately before the inflected verb or auxiliary, as in Germanic languages (Vikner 1995) or Dinka (van Urk 2015), whether the subject or another XP. Regular declarative clauses in Tira are topic clauses (Hassen et al. 2022): the initial constituent must be a DP, and this DP is indexed by an agreement prefix on the verb or auxiliary. Sentence-initial topics can be freely dropped, though their noun class is still indexed by topic agreement. For clarity, we will underline topics and topic agreement prefixes throughout the paper, and mark topic agreement with 'T' to distinguish it from other agreement suffixes that appear on the same head. The examples below illustrate a subject topic (1a) and an object topic (1b). The examples below are presented first in standard Tira orthography, then IPA.

(1) a. Ttuli ttamorza aprinya nddoba.
t̪òlé t̪-á àmòrð-à àprí-ɲá nd̪ɔ̀bà
CLt̪.lion CLt̪.T-AUX bite-FV CLj.boy-ACC tomorrow
'The lion is going to go bite the boy tomorrow.' Itive Imperfective

b. (Apri) ya ttuli amorza nddoba.
(àprí) j-á t̪òlé ámòrð-à nd̪ɔ̀bà
CLj.boy CLj.T-AUX CLt̪.lion bite-FV tomorrow
'(The boy,) the lion is going to go bite him tomorrow.' Itive Imperfective

* Himidan Hassen, University of Guelph, Peter Jenks, UC Berkeley. We are grateful to the audience at WCCFL43, Amy Rose Deal, and especially Nina Hagen Kaldhol and Sharon Rose for detailed comments as well as corrections of the Tira transcriptions. All remaining mistakes are entirely the fault of the second author.

When the subject is topicalized in (1a), the object *àprí-ɲá* ‘boy-ACC’ occurs after the verb and receives accusative case marking. The auxiliary *a* is phonologically bound to the verb when immediately preceding it as it triggers vowel hiatus resolution processes (Kaldhol 2024: 182–183); the auxiliary and verb are written as a single word when adjacent in Tira orthography. When the object is topicalized (1b), the subject *t̪òlé* ‘lion’ intervenes between the auxiliary and verb, and receives nominative case, which is unmarked.[1] Topics receive nominative, i.e., unmarked, regardless of whether they are underlying subjects or objects. We will show that topics are in [Spec, CP] and topic agreement occurs on C, while the post-auxiliary subject position is in [Spec, TP].

The examples in (1) are in the itive imperfective aspect/deixis inflection. This inflectional category is marked with a combination of morphological exponents: verb movement (only in the venitive perfective), the tone of the AUX vowel and verb, and the quality of an obligatory verb suffix that we call the ‘final vowel’ (FV). The itive perfective forms above, for example, have an L-toned auxiliary, a HL melody on the verb, and the final vowel is -ɛ̀. Itive imperfectives, on the other hand (e.g. (12b)), have a H-tone auxiliary, a HL tone melody on the verb, and a final -à suffix; see Kaldhol (2024: ch. 3) for a full discussion. To shorten our glosses we will simply indicate the inflectional category to the right of the gloss.

2.1. Topic agreement and oblique voice

In (1), the DP topic controls noun class agreement on the auxiliary. Noun class agreement on the verbal complex is restricted to topics in finite clauses. Topics can also be pronouns, but Tira makes minimal use of free pronouns, so such clauses have a sentence-initial auxiliary:

(2) a. Inggadongzatte Kukungu lebu kari unere.
íŋ-g-á dɔ́ŋð-àt̪-ɛ̀ kúkù-ŋú lə̀bú kárí únɛ́rɛ́
1SG-CLkT-AUX push-LOC.APPL-FV Kuku-ACC well inside yesterday
‘I pushed Kuku inside the well yesterday.’ Itive perfective

b. Anggadongzatte Kukungu lebu kari unere.
áŋ-g-à dɔ́ŋð-àt̪-ɛ̀ kúkù-ŋù lə̀bú kárí únɛ́rɛ́
2SG-CLkT-AUX push-LOC.APPL-FV Kuku-ACC well inside yesterday
‘You pushed Kuku inside the well yesterday.’ Itive perfective

Pronominal topics trigger person-number and noun class agreement on auxiliaries (see Section 2.3).

Oblique DPs can also be topics. Oblique topics trigger additional suffixes on the auxiliary that mark the semantic role of the oblique; we analyze these as a kind of ‘*wh*-agreement’ as they occur in focus and relative clauses as well, a form of oblique voice marking on C. There are two oblique *wh*-agreement markers. Locative DPs are marked with the suffix *-l* (3a). Locative *wh*-agreement is not a stranded adposition; in (3a) a separate adposition *kari* ‘inside’ is stranded in its base position (cf. (2a)). Instrumental DP topics are indexed by the suffix *-Cè* on the auxiliary, where *C* is class agreement (3b), so there is class agreement inside the instrumental *wh*-agreement marker.

(3) a. (Nganya) kal zondo idette kari unere.[2]
(ŋáɲà) k-à-l ðɔ̀ndɔ̀ ìd-ɛ̀t̪-ɛ̀ kárí únɛ́rɛ́.
CLk.bush CLk.T-AUX-WH:LOC CLð.gourd fall-APPL-FV inside yesterday
‘(The bush,) the gourd fell inside it yesterday.’ Itive perfective

b. (Zo) zaze apri koze Kukungu.
(ðô) ð-à-(ðɛ̀) àprí kɔ́ð-ɛ̀ kúkù-ŋù
CLð.stick CLð.T-AUX-WH:CLð.INST CLj.boy poke-FV CLk.Kuku-ACC
‘(The stick,) the boy poked Kuku away with it.’ Itive perfective

[1]The most reliable way to elicit sentences with natural topicalization of non-subjects is to ask for the translation of an English sentence where the target DP is the only 3SG pronoun in the clause. The corresponding Tira translation is most natural with the 3SG pronoun as the syntactic topic. Hence the parentheses on most non-subject topic examples.

[2]*ŋáɲà* ‘bush’ seems to control both *j*-class and *k*-class agreement in free alternation, e.g. /ŋáɲà jàl zɔ̀ndɔ̀ …/

Wh-agreement is a byproduct of topic agreement (Hassen et al. 2022), which we assume is *coarse*, i.e., it copies all available features on its goal (Baier 2018, Coon & Keine 2021, Deal & Royer 2025), including oblique locative and instrumental case values. In Hassen et al. (2022) we propose that topic agreement is driven by a composite probe on C which is only satisfied by D+[TOP]. This derives its ability to probe past non-topical DPs; see that paper for more evidence that the initial position is indeed an A′-position.

2.2. Pronoun agreement

We turn now to the second exponent of agreement on auxiliaries in Tira, pronoun agreement. As the name suggests, pronoun agreement is restricted to pronouns.[3] First, consider (4a), which has a subject topic and a pronominal object; the object pronoun is realized as a suffix on the auxiliary. Here and below, we will put boxes around exponents of pronoun agreement.

(4) a. Kuku ngganga dongzette nganya kari unere.

kúkù	ŋ-g-à-ŋâ	dɔ́ŋð-ɛ̀t̪-ɛ̀	ŋáɲà	kárí	únɛ́rɛ́
CLk.Kuku	3SG-CLk.T-AUX-2SG.O	push-LOC.APPL-FV	CLk.bush	inside	yesterday

‘Kuku pushed you into the bush yesterday.’ Itive Perfective

b. (Nganya) kangal Kuku dongzette kari unere.

(ŋáɲà)	k-à-ŋâ-l	kúkù	dɔ́ŋð-ɛ̀t̪-ɛ̀	kárí	únɛ́rɛ́
CLk.bush	CLk.T-AUX-2SG.O-LOC.WH	CLk.Kuku	push-LOC.APPL-FV	inside	yesterday

‘(The bush), Kuku pushed you into it yesterday.’ Itive Perfective

Importantly, pronoun agreement with the object occurs on the auxiliary regardless of the position of the subject; i.e. the overt DP subject in (4b) does not block pronoun agreement with the object.

Below we see examples of pronoun agreement with the subject, which occurs in cases of object or oblique topicalization with a pronominal subject. (Note that the vowel of the *a* ‘AUX’ deletes before or fuses with V-initial suffixes, we mark the missing morpheme in square brackets).

(5) a. (Kuku) ngge dongze.

(kúkù)	ŋ-g-ɛ̀	dɔ́ŋð-ɛ̀
CLk.Kuku	3SG-CLk.T-[AUX]-1SG.S	push-FV

‘(Kuku), I pushed him away.’ Itive Perfective

b. (Zo) zeze koze Kukungu.

(ðô)	ð-ɛ̀-ðɛ̀	kɔ̀ð-ɛ̀	kúkù-ŋú
CLð.stick	CLð.T-[AUX]-1SG.S-WH:CLð.INST	poke-FV	Kuku-ACC

‘(The stick,) I poked Kuku away with it.’ Itive Perfective

Subject pronoun agreement is distinct from object pronoun agreement:

(6) a. Kuku nggangi dongze.

kúkù	ŋ-g-à-ŋɛ̂	dɔ́ŋð-ɛ̀
CLk.Kuku	3SG-CLk.T-AUX-1SG.O	push-FV

‘Kuku pushed me away.’ Itive Perfective

b. (Zo) zangize Kuku koze.

(ðô)	ð-à-ŋɛ̂-ðɛ̀	kúkù	kɔ́ð-ɛ̀
CLð.stick	CLð.T-AUX-1SG.O-WH:CLð.INST	CLk.Kuku	poke-FV

‘(The stick,) Kuku poked me away with it.’ Itive perfective

[3] We are setting aside verbs in the venitive perfective, where V moves to T, then to C. These verb forms lack an auxiliary; topic agreement is a prefixes on the verb, while pronoun agreement and oblique voice are suffixes on the verb itself, now occupying the AUX position (see fn. 4). As predicted by a verb-movement account, in this verb form non-topical subjects must be postverbal.

Example (6b) shows that object agreement can skip a lexical DP subject: 3rd singular subject pronoun agreement is *-ŋ*, here absent. This observation supports the characterization of pronoun agreement as sensitive only to pronouns. Below we model this aspect of pronoun agreement with the proposal that only pronouns possess a π-feature (from Béjar 2003), absent on regular lexical DPs.

2.3. Two structural sources of agreement

There are several differences between topic agreement and pronoun agreement which suggest they are exponents of distinct probes. The first difference is in their targets of agreement: Topic agreement can index any DP topic, whether a pronoun, proper noun, or noun phrase, pronoun agreement only indexes pronominal arguments.

Second, only topic agreement ever indexes noun class features.[4]

(7) Agreement in Tira finite clauses

	T(opic agreement)	S(ubject pronoun)	O(bject pronoun)
DP	CL-	none	none
1SG	íŋ-g-	-ɛ̀	-ŋɛ̂
2SG	á-g-	-á	-ŋâ
3SG.HUM	ŋ-g-	-ŋ	-∅
1IN.DU	á-l-	-l	-t̪ɛ́
1IN.PL	á-l -r	-l -r	-t̪ɛ́ -r
1EX.PL	ɲá-l-	-íɲà	-éɲár
2PL	ɲá-l-	-íɲá	-tɛ́
3PL.HUM	l-	-ɪ́	-l(í)

Topic agreement for pronominal topics indexes both person and class g/l noun class features: singular person prefixes are followed by a *g*-, a voiced allophone of singular *k*-class agreement, and plural *l*-class agreement for plural person markers. So the third person topic prefix *ŋg*- includes two components: *ŋ*- for third person and *g*- as the voiced *k*-class agreement. Nonhuman 3rd person subjects only every occur with class agreement; see example (3a) where *k*-class *nganya* 'bush' is tracked only with class agreement. In fact, the 3SG *ŋ*- prefix is often deemed optional with human subjects, which can occur with only class agreement, and must do so for non *k*-class human subjects.

The third difference is in the DPs they interact with and how many they will seek. While sentences only have a single DP topic, pronoun agreement can look past full DP arguments to agree with all of the pronouns in the clause, as we will see in the following section on multiple pronoun agreement.

Fourth, the morphological position of the two kinds of agreement is distinct. While topic agreement is realized as a prefix or final suffix, pronoun agreement is realized as a suffix directly attached to the auxiliary vowel *a*. These ordering facts follow from the proposal that pronoun agreement is on T, topic agreement is on C, and T always moves to C in Tira. Hence, the ordering facts reduce to an instance of the mirror principle: pronoun agreement is inside of topic agreement because it is lower on the tree:

(8) $[\text{ Top-[AUX-S-O]}_T\text{-Top}_{wh}]_C$

The crucial ordering fact is that the oblique wh-suffixes generally follow pronoun agreement suffixes, consistent with the mirror principle.

A syntactic argument that topic agreement is higher than pronoun agreement comes from infinitive clausal complements. In such clauses, topic agreement is absent while pronoun agreement still occurs,

[4]The exception is verbs in the venitive perfective, which feature verb movement (fn. 3), and can topic agree for class or person but never double the two:

(1) Kuku kedongzengi. (cf. *Kuku nggedongzengi)
<u>kúkù</u> <u>k</u>ə̀-dɔ̀ŋð-ɛ́ŋì
CLk.Kuku CLk.T-push-1SG.O
'Kuku pushed me away.' Venitive Perfective

Note person agreement in this form is suffixed directly on the verb, due to verb movement.

though as a verbal prefix. Infinitive clauses also lack topics. These facts follow if infinitival clauses lack CP, hence both topics and topic agreement. Nevertheless, infinitives have a paradigm resembling pronoun agreement: they do not agree for noun class, but they do have regular 3rd person singular agreement. Witness the raising-to-object constructions below:

(9) a. [$_{CP}$ Kuku nggaytte ttulinya$_i$ [$_{TP}$ t_i ngehinya diyo.]]

kúkù ŋg-à áít̪-ὲ t̪òlè-ɲá ŋə̀-ɽíɲ-à dìjɔ́
CLk.Kuku 3SG.CLk.T-AUX allow-FV CLt̪.lion-ACC 3SG.S-kill-INF cow.ACC

'Kuku allowed the lion to kill the cow.' Itive perfective

b. [$_{CP}$ Ttuli$_i$ tta ayttine t_i [$_{TP}$ t_i ngehinya diyo]].

t̪òlé$_i$ t̪-à áít̪-ìn-ὲ t_i ŋə̀-ɽíɲ-à dìjɔ́
CLt̪.lion CLt̪.T-AUX allow-PASS-FV 3SG.S-kill-INF cow.ACC

'The lion was allowed to kill the cow.' Itive perfective

In (9a), the main clause shows *k*-class topic agreement with the topicalized subject Kuku, while the embedded clause shows only 3SG infinitive subject agreement *ŋ(ə)-*. In (9b) the raised object is passivized and indexed by *t̪*-class topic agreement; however, infinitive 3SG *ŋ-* agreement remains in the lower clause, never class agreement. This difference follows from the idea that class agreement is restricted to finite C, while person agreement is typical of T heads, with *ŋ-* prefixes typical of infinitive T.

Interestingly, topic agreement is never doubled by subject pronoun agreement in main clauses:

(10) * íŋ-g-ὲ və́lὲð-ὲ ðàŋàl-à
1SG-CLg.T-[AUX]-1SG.S pull-FV CLð.sheep-ACC

This restriction is an instance of Kinyalolo's Constraint (Kinyalolo 1991, Carstens 2005, Baker 2010):

(11) Kinyalolo's Constraint (as stated by Baker 2010)
*Agreement on a lower head with NP X if there is a higher head that also agrees with X and the two functional heads are part of the same word at PF.

In short, we have seen that there is good evidence for the model of Tira clauses in Figure 1, where subjects are in [Spec, TP], topics are in [Spec, CP], T agrees with all pronouns, C agrees with the topic, and T always moves to C.

DP subjects surface in [Spec, TP] despite the fact that T does not agree with them. Hence, pronoun agreement must be distinct from a regular EPP-type subject requirement. Evidence that the subject position in Tira is [Spec, TP] rather than a lower thematic position comes from standard A-movement phenomena. For example, the passivized subject *ðɔ́ndɔ̀* in (12b):

(12) a. (Apri) ya Kuku ngaci zonda unere.

(àprí) j-à kúkù ŋàc-ì ðɔ̀nd-à únɛ́rɛ́
(CLj.boy) CLj-AUX CLk.Kuku give-FV CLð.gourd-ACC yesterday'

'(The boy,) Kuku gave the gourd to him yesterday' (far) Itive Perfective

b. (Apri) ya zondo ngacino nddoba.

(àprí) j-á ðɔ́ndɔ̀ ŋàc-ìn-ɔ̀ ndɔ̀bà
CLj.boy CLj-AUX CLð.gourd give-PASS-FV tomorrow

'(The boy,) the gourd is going be given to him tomorrow' (far) Itive Imperfective

Passivization thus must involve movement of the object to a regular derived subject position between the auxiliary and before the lexical verb, by hypothesis, [Spec, TP].

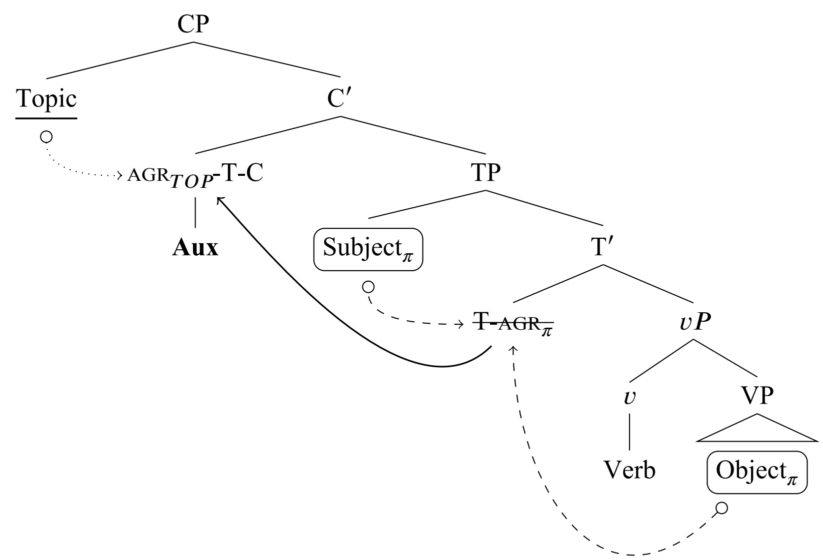

Figure 1: Tira topic agreement on C, pronoun agreement on T, and T-to-C movement

3. Multiple pronoun agreement and Expone Outermost

Agreement patterns in Tira are especially interesting when there are at least three eligible targets for agreement. If the topic is an oblique, for example, and both the subject and object are pronouns, then all three arguments can be indexed on the auxiliary in finite clauses. Most examples, like 13a, follow the template in (8), though there are occasional deviations. For example, in (13b), locative wh-agreement *-l* occurs inside the 1IN.DU object marker t̪ɛ́.

(13) a. (Zupi) zangteze koze. 3sg>2pl
(ðùpì) ð̠-à-ŋ-t̪ɛ́-ðɛ́ kɔ́ð-ɛ̀.
CLð.stick CLð.T-AUX-3SG.S-2PL.O-INST.WH poke-FV
'(The stick,) s/he poked you all away with it.' Itive perfective

b. (Lebu) langalte dongzette kari unere. 3sg>1in.du
(lə̀bú) l̠-à-ŋ-l̠-t̪ɛ́ dɔ́ŋð-ɛ́t̪-ɛ́ kárí únɛ́rɛ́
CLl.well CLl.T-AUX-3SG.S-LOC.WH-1IN.DU.O push-LOC-FV inside yesterday
'(The well,) s/he pushed me and you into it yesterday.' Itive perfective

These examples show that pronominal agreement is *insatiable*, agreeing with all pronouns in its clause.

We model insatiability with the interaction/satisfaction framework (Deal 2024), assuming the person agreement probe on T has the following features:

(14) [int: π, sat: -]

The interaction condition is π, which only occurs on pronouns (Béjar 2003, Sichel & Toosarvandani 2025, a.o.). The satisfaction condition is null, producing insatiability.

We turn now to our central puzzle, which consists of occasional gaps in the realization of multiple pronoun agreement. In certain combinations of pronominal subjects/objects, only object φ-features are realized, resulting in subject ambiguity:

(15) a. (Zo) zangize koze. 2sgS/3sgS>1sgO
(ðô) ð̠-à-ŋɛ̂-ðɛ̀ kɔ̀ð-ɛ̀
CLð.stick CLð-AUX-1SG.O-WH:CLð.INST poke-FV
'(The stick,) you/she poked me away with it.' Itive perfective

b. (Zo) zangaze koze. 1sgS/3sgS>2sgO

(ðô) ð-à-ŋâ-ðɛ̀ kɔ̀ð-ɛ̀
CLð.stick CLð-AUX-2SG.O-WH:CLð.INST poke-FV

'(The stick,) I/she poked you away with it.' Itive perfective

c. (Zo) zaze koze. 2sgS/3sgS>3sgO

(ðô) ð-à-∅-ðɛ̀ kɔ̀ð-ɛ̀
CLð.stick CLð-AUX-3SG.O-WH:CLð.INST poke-FV

'(The stick,) you/she poked him/her away with it.' Itive perfective

In each of these examples, spontaneously produced by the first author in response to a translation task on multiple days, subject prefixes are unexpectedly missing. These examples form a natural class in that they all involve multiple singular arguments; only 1sgS>3sgO realizes subject features reliably, though 3SG object agreement is null:

(16) (Zo) zeze koze. 1sgS>3sgO

(ðô) ð-ɛ̀-∅-ðɛ̀ kɔ́ð-ɛ̀
CLð.stick CLð-[AUX]-1SG.S-3SG.O-WH:CLð.INST poke-FV

'(The stick,) I poked him/her away with it.' Itive perfective

So perhaps (15c) should be analyzed with an overt 2sgS *-à*. Still, 3sgS *-ŋ* is always unexpectedly absent.

Beyond the SGS>SGO forms, object-only realizations of multiple pronominal agreement also occur with 1st person exclusive plural objects, with the exception of with third person plural subjects:

(17) a. (Zo) zenyarze koze. 2sgS/3sgS/2plS>1ex.plO

(ðô) ð-éɲár-ðɛ́ kɔ́ð-ɛ̀
CLð.stick CLð-[AUX]-1EX.PLO-WH:CLð.INST poke-FV

'(The stick,) you/he/y'all poked us (not you) away with it.' Itive perfective

b. (Zo) zenyarlze koze. 3plS>1ex.plO

(ðô) ð-éɲár-l-ðɛ́ kɔ́ð-ɛ̀
CLð.stick CLð-[AUX]-1EX.PLO-3PLS-WH:CLð.INST poke-FV

'(The stick,) they poked us (not you) away with it.' Itive perfective

Table 1 summarizes where pronoun agreement surfaces for both arguments versus just the object; we present these cases schematically, and will include complete paradigms in future work.

Could phonological explanations account for some of the apparent gaps? For example, 3SGS agreement, *-ŋ*, might be deleted before the palatal nasal in the following syllable, or the 2SGS *-á* might be fronted to [e] before the following palatal nasal. However, the sequence [ŋaɲ] is well-formed in Tira, as the word [ŋáɲà] 'bush' demonstrates, rendering a phonological account implausible.

Subj↓/Obj→	1SG	2SG	3SG	1IN.DU	1IN.PL	1EX	2PL	3PL
1SG	-	(S-)O	S-O		-	-	S-O	S-O
2SG	O	-	O	-	-	O	-	S-O
3SG	O	O	O	S-O	S-O	O	S-O	S-O
1IN.DU	-	-	S-O	-	-	-	-	S-O
1IN.PL	-	-	S-O	-	-	-	-	S-O
1EX	-	S-O	S-O	-	-	-	S-O	S/O
2PL	S-O	-	S-O	-	-	O	-	S-O
3PL	S-O	S-O	S-O	S-O	S-O	S-O	S-O	S-O

Table 1: Realization of subject and object in multiple pronoun agreement configurations (NB. Empty cells are impossible or reflexive)

3.1. *Expone outermost as order-sensitive impoverishment*

What explains these gaps? An initially appealing approach to such cases is as an object preference. For example, some instances of verbal agreement preferentially agree with an object given certain feature specifications, and often fail to track subject features altogether once these preferences are satisfied; as in cases successfully modeled by Cyclic Agree (Béjar & Rezac 2009, Clem 2022). However, this approach makes several incorrect predictions about Tira. First, pronoun agreement is on T, so there is no way to have a probe on T the object before the subject; if anything we expect a subject preference. Additionally, we do not find any person hierarchy effects for pronoun agreement, counter the expectations of a Cyclic Agree approach. For example, 3SG subjects are successfully copied in the context of 1st inclusive dual objects (13b), while 1SG subjects can be copied alongside 3SG objects (16). If, for example, first person features halted probing, we would expect only partial agreement in one direction or the other.

Instead, the object preferences seem like an instance of Expone Outermost, when a multiply-valued probe only realizes the features of the final goal which it is valued by (Grishin to appear):

(18) EXPONE OUTERMOST
On a multiply-valued head $H_{([A], [B],\ldots[Z])}$, expone only the outermost feature bundle [Z]: $H_{[Z]}$.

Grishin, following Deal (2022b), examines several instances of Expone Outermost, the simplest of which involve case-overwriting, where a DP is assigned two case values over the course of a derivation but only the last surfaces.

In fact, case-overwriting is common in Tira. Consider SVO/OVS alternations: when objects, which are always accusative, move to topic position, they surface with nominative case:

(19) a. Ttuli ttamorza aprinya nddoba.
t̪òlé t̪-á ámòrð-à àprí-ɲá nd̪ɔ̀bà
CLt̪.lion CLt̪.T-AUX bite-FV CLj.boy-ACC tomorrow
'The lion is going to go bite the boy tomorrow.' Itive Imperfective

b. Aprí ya ttuli amorza nddoba.
àprí j-á t̪ólé àmòrð-à nd̪ɔ̀bà
CLj.boy CLj.T-AUX CLt̪.lion bite-FV tomorrow
'(The boy,) the lion is going to go bite him tomorrow.' Itive Imperfective

This pattern presumably involves multiple case assignment to *àprí* 'boy', resulting in a morphological representation where this DPs case features consist of a tuple isomorphic to its derivational history (Deal 2015, Hammerly 2020):

(20) /àprí(-ɲá)/: ($n_{[CLj,SG]}$, ACC, NOM)

So some preference—Expone Outermost—produces surface forms with only nominative case.

One way of thinking about Expone Outermost is as a parameter on probes or languages that apply across-the-board. Tira person agreement seems to pose a puzzle for this conception of Expone Outermost: while most combinations of subject and object surface faithfully (13), in other contexts just the features of the subject are deleted.

It seems instead that Expone Outermost is an instance of order-sensitive impoverishment—the contextual deletion of specific morphological features (Bonet 1991, Baier 2018, Keine & Müller to appear: a.o.)—but one which makes reference to an ordered set or tuple:

(21) Expone Outermost as order-sensitive impoverishment
a. Given a multiply-valued head $H_{([F_1], [F_2],\ldots, [F_n])}$
b. $F_n \rightarrow \varnothing / \_F_{n\pm1}$

This Deal (2022a), to account for an apparent case of Expone Innermost in Khanty, which Colley & Privoznov (2020) argue involves partial agreement by a topic probe—one which exists along with Expone Outermost on the same probe.

We apply this to a simple case like the deletion of an accusative in the context of nominative. A rule like the following would suffice (where CASE is a cover term for morphological case features):

(22) Tira order-sensitive case impoverishment

a. $\text{CASE}_n \rightarrow \varnothing/\_\text{CASE}_{n+1}$

b. *ṱòlé*: ($n_{[CLṱ,SG]}$, ACC, NOM) → *ṱòlé*: ($n_{[CLṱ,SG]}$, NOM)

This brings us back to the cases in (15) and (17a), summarized in Table 1.

The resulting analysis is unattractive but accounts for the data, as the cases can be captured by a hodgepodge of four rules, written below using the feature system for person of Harbour (2016):

(23) Tira order-sensitive impoverishment of person agreement features

a. $[-\text{SPEAKER,SG}]_n \rightarrow \varnothing/\_[\text{SG}]_{n+1}$ (delete 2SG and 3SG subjects before SG objects)

b. $[+\text{SPEAKER,SG}]_n \rightarrow \varnothing/\_[+\text{PART,SG}]_{n+1}$ (delete 1SG subjects before 2SG objects)

c. $[+\text{PART}]_n \rightarrow \varnothing/\_[+\text{SPEAKER,PL}]_{n+1}$ (delete 2SG/PL subjects before 1EX.PL objects)

d. $[-\text{PART,SG}]_n \rightarrow \varnothing/\_[-\text{SPEAKER,PL}]_{n+1}$ (delete 3SG subjects before 1EX.PL objects)

The rule in (23b) is optional; while only impoverished examples have been produced in spontaneous speech, the first person pronoun was accepted when offered as an alternative:

(24) Free alternates in 1SG>2SG contexts

a. /ð-à-ŋâ-ðɛ́/ CLð.T-AUX-2SG.O-WH:CLð.INST

b. /ð-ɛ̀-ŋâ-ðɛ́/ CLðT-1SGS-[AUX]-2SGO-WH:CLð.INST

Thus, by modeling Expone Outermost as order-sensitive contextual impoverishment, we can capture the sporadic non-occurrence of multiple subject agreement in Tira.

Ditransitives provide additional support for this analysis, though not without complications. Support comes from the fact that the same patterns of multiple agreement and Expone Outermost occur between subjects and objects when these objects are recipients (25) in sentences with a topicalized theme:

(25) a. (Zondo) zanga ngaci.

(ðɔ̀ndɔ̀) ð-à-ŋâ ŋátʃ-ì.
CLð.gourd CLð.T-AUX-2SG.O give-FV

'(The gourd,) I/she gave it to you.' Itive Perfective

b. (Zondo) zangi ngaci.

(ðɔ̀ndɔ̀) ð-à-ŋɛ̂ ŋátʃ-ì.
CLð.gourd CLð.T-AUX-1SG.O give-FV

'(The gourd,) you/she gave it to me.' Itive Perfective

The form *ðéŋá*, with an overt 1SG subject agreement suffix *-é* is also allowed for (25a), consistent with the idea that the 1sgS deletion rule is optional in this context.

There is only one object pronoun slot available in Tira, which raises the question of what happens when both objects are non-topic pronouns. First, observe that recipients must precede themes in Tira when both are noun phrases (26a). As predicted, then, object pronoun agreement when both are pronouns is with the theme (26b):

(26) a. Kuku kangci aprinya urno

kúkù k-àŋtʃ-í áprì-ɲá úrnò
CLk.Kuku CLk.T-[AUX]-show-FV boy-ACC grandfather.ACC

'Kuku showed the boy to the grandfather.' (*'the grandfather to the boy') Itive Perfective

b. Kuku kanga angci ngung/nging.

kúkù	k-à-ŋâ	àŋtʃ-í	(ŋú-ŋ)/(ŋí-ŋ)
CLk.Kuku	CLk.T-AUX-2SG.O	show-FV	3SG-ACC/1SG-ACC

'Kuku showed you to him/me.' Itive Perfective

The theme-only pattern in (26b) seems to be another instance of Expone Outermost. One complication is that such sentences can include a repair where the missing recipient is pronounced as a free pronominal object. Another complication is that sentences like (26b) were sometimes judged to be acceptable translations of sentences with second person recipients. This suggests that deletion might apply in either order in the case of multiple objects. Here more work is needed.

In conclusion, we have seen that agreement in Tira finite clauses is due to two separate probes, one on C and another on T. T-to-C head movement places both probes on a single head. This sets the stage for deletion operations, including Kinyalolo's Constraint, which deletes features on T when they are identical to those on C, and Expone Outermost effects, which seems to intermittently delete subject pronoun features in the presence of object pronoun features. Because these rules need to occur only in specific morphological combinations, Expone Outermost patterns perhaps should be modeled as an order-sensitive impoverishment operation.

References

Baier, Nico. 2018. *Anti-agreement*. UC Berkeley dissertation.

Baker, Mark C. 2010. On parameters of agreement in Austronesian languages. In *Austronesian and theoretical linguistics*, 345–374. John Benjamins Publishing Company.

Béjar, Susana. 2003. *Phi-syntax: a theory of agreement*. University of Toronto dissertation.

Béjar, Susana & Milan Rezac. 2009. Cyclic agree. *Linguistic Inquiry* 40(1). 35–73.

Bonet, Eulalia. 1991. *Morphology after syntax–pronominal clitics in Romance*. MIT dissertation.

Carstens, Vicki. 2005. Agree and epp in bantu. *Natural Language & Linguistic Theory* 23(2). 219–279.

Clem, Emily. 2022. Cyclic expansion in Agree: maximal projections as probes. *Linguistic Inquiry* 54. 39–78.

Colley, Justin & Dmitry Privoznov. 2020. On the topic of subjects: Composite probes in Khanty. In *Proceedings of NELS*, vol. 50, 111–124.

Coon, Jessica & Stefan Keine. 2021. Feature gluttony. *Linguistic inquiry* 52(4). 655–710.

Deal, Amy Rose. 2015. Interaction and satisfaction in φ-agreement. In *Proceedings of nels*, vol. 45, 179–192.

Deal, Amy Rose. 2022a. Composite probes. *Handout for course at MIT*.

Deal, Amy Rose. 2022b. Systems of ϕ-agreement. *Handout for course at MIT*.

Deal, Amy Rose. 2024. Interaction, satisfaction, and the PCC. *Linguistic Inquiry*. 1–56.

Deal, Amy Rose & Justin Royer. 2025. Mayan animacy hierarchy effects and the dynamics of Agree. *Natural Language & Linguistic Theory*. 1–57.

Grishin, Peter. to appear. How to agree with the lowest DP. *Natural Language & Linguistic Theory*.

Hammerly, Christopher. 2020. *Person-based prominence in Ojibwe*. U. of Massachusetts, Amherst dissertation.

Harbour, Daniel. 2016. *Impossible persons*. Vol. 74. MIT Press.

Hassen, Himidan, Peter Jenks & Sharon Rose. 2022. A′-satisfaction with φ-interaction in Tira. Presentation at WCCFL 40, Stanford University.

Kaldhol, Nina Hagen. 2024. *Tonal dimensions of morphological complexity*. UC San Diego dissertation.

Kaldhol, Nina Hagen, Sharon Rose & Mark Simmons. 2025. Prosody of topic and focus in Tira. In A. Akinlabi, S. Korsah, A. R. Sulemana & S. Rose (eds.), *Cross-disciplinary approaches to information structure in Niger-Congo languages*, 51–82. Berlin: Language Sciences Press.

Keine, Stefan & Gereon Müller. to appear. Impoverishment. In A Alexiadou, R. Kramer, A. Marantz & I. Oltra-Massuet (eds.), *The Cambridge handbook of distributed morphology*. CUP.

Kinyalolo, Kasangati Kikuni Wabongambilu. 1991. *Syntactic dependencies and the spec-head agreement hypothesis in Kilega*. UCLA dissertation.

Rose, Sharon. 2025. Inalienable personal possessives in Moro and Tira. In H. F. Idris, J. Junglas & G. Schneider-Blum (eds.), *Nuba Mountain language studies – contemporary perspectives*, 149–174. University of Cologne: USB Monographs.

Sichel, Ivy & Maziar Toosarvandani. 2025. The featural life of nominals. *Linguistic Inquiry*. 751–806.

van Urk, Coppe. 2015. *A uniform syntax for phrasal movement: a case study of Dinka Bor*. MIT dissertation.

Vikner, Sven. 1995. *Verb movement and expletive subjects in the Germanic languages*. Oxford University Press.